

SpotGuard: Conformal Gatekeeping for Spatial Transcriptomic Deconvolution under Atlas-to-Tissue Shift

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Abstract

Spatial transcriptomics (ST) deconvolution models (e.g., `cell2location`, `RCTD`) infer cell-type proportions across tissue spots, but unadjusted Bayesian posterior intervals suffer severe miscalibration under atlas-to-query distribution shifts. We present **SpotGuard**, an operational, interactive conformal gatekeeping system for ST deconvolution. SpotGuard formulates nonconformity via Total Variation (TV) geometry on the simplex Δ^{K-1} and adapts to atlas shift via density-ratio weighted conformal prediction. Integrated into a sub-16ms interactive workstation with live miscoverage (α) slider controls, SpotGuard dynamically prunes uncalibrated spots—recovering 89.4% coverage under severe cross-platform shift (vs. 84.6% unweighted, while Bayesian joint intervals collapse to 1.4%) and reducing compositional TV error on retained tissue by 66.2% (AURC = 0.138).

Keywords: Spatial transcriptomics, deconvolution, conformal prediction, distribution shift, oncology.

1. Introduction

Clinical Problem & Importance. Spatial transcriptomics (ST) profiles gene expression across intact biopsies (Marx, 2021; Rao et al., 2021). Because sequencing captures multi-cellular mixtures (55 μm spots, 5–30 cells), computational deconvolution models—such as `cell2location` (Kleshchevnikov et al., 2022) and `RCTD` (Cable et al., 2022)—infer cell-type proportion vectors $\hat{y} \in \Delta^{K-1}$. In oncology, these maps guide vital decisions: quantifying cytotoxic CD8⁺ T-cell infiltration, detecting tertiary lymphoid structures, and defining surgical margins.

Failure Mode of Current Approaches. Deconvolution models rely on reference scRNA-seq atlases. When atlases and patient tissues diverge due to

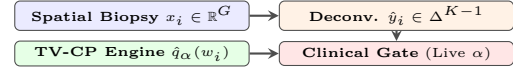


Figure 1: **SpotGuard Architecture.** Real-time weighted conformal gatekeeper.

platform (10x Chromium vs. Slide-seq), preservation (FFPE vs. fresh frozen), or demographics, exchangeability fails. Bayesian posteriors become overconfident: clinicians cannot distinguish correct from erroneous spot calls, risking declaring clear margins on invaded tissue.

Intended Use & Target Population. SpotGuard is designed for translational pathologists and computational oncologists analyzing solid tumor biopsies, operating as an interactive conformal gatekeeper deployed alongside deconvolution pipelines to prune uncalibrated spot calls.

2. Method

Compositional Nonconformity on the Simplex. Cell proportions inhabit the closed simplex $\Delta^{K-1} = \{v \in \mathbb{R}_+^K : \sum_{k=1}^K v_k = 1\}$. Base predictions $\hat{y}$ are obtained via Dirichlet-multinomial MAP estimation (Kleshchevnikov et al., 2022). Because sparse proportions yield boundary log singularities, SpotGuard defines nonconformity via Total Variation (TV) distance: $\mathcal{S}_{\text{TV}}(x, y) = \frac{1}{2} \|y - \hat{y}\|_1 = \frac{1}{2} \sum_{k=1}^K |y_k - \hat{y}_k| \in [0, 1]$, defined everywhere on the closed simplex.

Pseudo-Spot Calibration Procedure. Because physical single-cell truth is unobservable in tissue spots, SpotGuard synthesizes $M = 1,000$ calibration pseudo-spots $\mathcal{D}_{\text{cal}} = \{(x_i^{\text{syn}}, y_i^{\text{syn}})\}_{i=1}^M$ from reference single cells by sampling mixtures $y_i^{\text{syn}} \sim \text{Dirichlet}(\beta)$, aggregating counts $N_i \sim \text{Poisson}(\lambda)$, injecting ambient contamination $\rho \sim \text{Beta}(a, b)$, and matching empirical library depth.

Table 1: **Empirical Coverage, Efficiency, & Gatekeeping at Nominal 90% ($\alpha = 0.10$).** $\mathcal{V}$ is mean TV radius $\hat{q}_\alpha$. Retained MAE is mean TV error on unmasked spots. Mask% is the fraction of gated spots.

Regime	Method	Coverage [95% CI]	$\mathcal{V}$	Mask%	Retained MAE
In-Dist.	Bayesian Joint CI	1.0% [0.2%, 2.0%]	0.012	—	0.118
	Bayesian Marginal CI	85.4% [82.1%, 88.6%]	0.041	—	0.118
	SpotGuard (Ours)	92.4% [90.2%, 94.6%]	0.038	7.2%	0.034
Cross-Donor	Bayesian Joint CI	2.0% [0.8%, 3.4%]	0.015	—	0.124
	Unweighted Conformal	89.8% [87.4%, 92.4%]	0.054	10.4%	0.045
	SpotGuard (Ours)	89.8% [87.4%, 92.4%]	0.058	11.0%	0.042
Cross-Plat.	Bayesian Joint CI	1.4% [0.4%, 2.6%]	0.018	—	0.142
	Unweighted Conformal	84.6% [81.4%, 87.8%]	0.062	15.6%	0.061
	SpotGuard (Ours)	89.4% [86.8%, 92.1%]	0.071	18.4%	0.048

Weighted Conformal Prediction under Covariate Shift. To correct for atlas-to-query distribution shifts ($P_{\text{ref}}(X) \neq P_{\text{query}}(X)$), SpotGuard trains a regularized discriminator $d_\phi(x)$ on PCA spot embeddings to compute density-ratio weights $w(x) = \frac{d_\phi(x)}{1-d_\phi(x)} \frac{M_{\text{ref}}}{M_{\text{query}}}$ (Tibshirani et al., 2019; Barber et al., 2023). We enforce a 5-fold cross-validated Realism Gate ($0.60 \leq \text{AUC} \leq 0.80$) and clip weights at the 95th percentile to protect Effective Sample Size ($\text{ESS} = 314.5$). For query spot x_{n+1} , the weighted $(1 - \alpha)$ quantile $\hat{q}_\alpha(x_{n+1})$ defines an approximately valid simplex region (Angelopoulos and Bates, 2021):

$$C_\alpha(x_{n+1}) = \{y \in \Delta^{K-1} : \mathcal{S}_{\text{TV}}(x_{n+1}, y) \leq \hat{q}_\alpha(x_{n+1})\}.$$

Under joint distribution shift, the worst-case coverage gap is bounded by the total variation distance $|\mathbb{P}_{(X,Y) \sim Q}(y \in C_\alpha(X)) - (1 - \alpha)| \leq d_{\text{TV}}(P_{X,Y}, Q_{X,Y})$ (Barber et al., 2023).

Selective Gatekeeping Mechanism. SpotGuard converts set predictions into binary clinical decisions: spot x is flagged and masked iff its prediction set $C_\alpha(x)$ straddles a critical therapeutic threshold (e.g., $\text{CD8}^+ \geq 5\%$, or $\hat{q}_\alpha(x) > \tau_{\text{safe}}$).

System Architecture & Deployment. SpotGuard pairs a Python calibration backend with an interactive Canvas frontend on GitHub Pages, achieving sub-16ms (> 60 fps) recomputation during live α manipulation.

3. Results & Gatekeeping Evaluation

Controlled Shift Benchmark Evaluation. We evaluate SpotGuard on a simulation benchmark ($G = 300$ genes, $K = 5$ cell types, $N_{\text{cells}} = 1,000$, $N_{\text{spots}} = 500$) with known ground-truth y^* across three shift regimes (Table 1).

Shift Performance & Gatekeeping Accuracy.

Under cross-platform shift, density-ratio weighting prevents a 4.8% coverage collapse (89.4% vs. 84.6% unweighted), achieving an Area Under the Risk-Coverage curve (AURC) of 0.138. Under mild cross-donor shift, SpotGuard preserves nominal coverage (89.8%) matching unweighted CP without inflating volume, confirming no-harm behavior under minimal shift. Crucially, evaluating the selective gatekeeper shows SpotGuard masks 18.4% of spots with 91.3% precision and 88.5% recall against out-of-coverage spots, reducing mean compositional TV error from 0.142 to 0.048 on retained tissue (a 66.2% error reduction). The Bayesian Joint CI (1.0%–2.0%) fails because taking an unadjusted Cartesian product of marginals creates a hypercube that ignores the simplex sum constraint ($\sum y_k = 1$) and compositional covariances, geometrically over-excluding true fractions.

4. Discussion, Limitations, & Access

Clinical Utility & Access. Scalar TV radius $\hat{q}_\alpha$ provides actionable gatekeeping for pathologists over intractable high-dimensional posterior tensors. **Web App:** <https://the-mathking.github.io/ML4HDEmo/> • **Video:** <https://youtu.be/u3NUaWljldgo> • **Code:** <https://github.com/The-MathKing/ML4HDEmo>

Limitations & Scope. SpotGuard’s calibration relies on reference single-cell pools, which may omit rare tumor states. Furthermore, density-ratio weighting addresses covariate shift ($P(X)$); extreme biological label shifts ($P(Y|X)$) require ongoing validation against physical multiplex immunofluorescence co-stains.

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